

SOME SOLUTIONS TO THE PLANETARY GEOSTROPHIC EQUATIONS

TWNH

Peterson and Callies (in review) (see also *Peterson and Callies* 2023) consider how bottom-intensified vertical mixing induces a basin-scale circulation in various simple non-flat-bottomed basins. They write equations for a bottom Ekman layer and couple them to an equation for the geostrophic contour vorticity balance, in the parlance of *Waldman and Giordani* (2023), which is a type of barotropic vorticity equation.

1. GOVERNING EQUATIONS

The flow satisfies the (non-dimensional) planetary-geostrophic equations:

$$(1) \quad -fv = -\frac{\partial p}{\partial x} + \epsilon^2 \frac{\partial}{\partial z} \left(\nu \frac{\partial u}{\partial z} \right),$$

$$(2) \quad fu = -\frac{\partial p}{\partial y} + \epsilon^2 \frac{\partial}{\partial z} \left(\nu \frac{\partial v}{\partial z} \right),$$

(3)

(see Eqs. (9)–(11) of *Peterson and Callies* in review, Eq. (3.26) of *Klinger and Haine* 2019, and Eq. (2.274) of *Vallis* 2006). The boundary conditions are no flow at the bottom and a specified stress at the surface:

$$(4) \quad \text{Surface } z = 0 : \epsilon^2 \nu \partial_z(u, v) = (\tau^x, \tau^y),$$

$$(5) \quad \text{Bottom } z = -H : (u, v) = 0,$$

The flow is non-divergent:

$$(6) \quad \nabla \cdot \mathbf{u} = 0,$$

the buoyancy field is hydrostatic:

$$(7) \quad \frac{\partial p}{\partial z} = b,$$

and the buoyancy field satisfies the advection-diffusion-relaxation equation

$$(8) \quad \mu \rho \left(\frac{\partial b}{\partial t} + \mathbf{u} \cdot \nabla b \right) = \epsilon^2 \frac{\partial}{\partial z} \left(\kappa \frac{\partial b}{\partial z} \right) + \gamma (B - b).$$

(see Eqs. (11)–(13) of *Peterson and Callies* in review). The boundary conditions on buoyancy are:

$$(9) \quad \text{Surface } z = 0 : b = 0$$

$$(10) \quad \text{Bottom } z = -H : \frac{\partial b}{\partial z} = 0,$$

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and the initial condition is $b = z$. The buoyancy field relaxes to a specified field $B(x, y, z, t)$ with rate γ .

The diffusivity and viscosity are given by

$$(11) \quad \kappa(z) = \kappa_0 + \exp(-\alpha(z + H)),$$

$$(12) \quad \nu = \text{constant}$$

Peterson and Callies (in review) Eq. (17).

2. VERTICALLY-INTEGRATED FLOW

The vertically-integrated flow (U, V) is given by:

$$(13) \quad \begin{aligned} U &\equiv \int_{-H}^0 u \, dz = - \int_{-H}^0 z \partial_z u \, dz, \\ V &\equiv \int_{-H}^0 v \, dz = - \int_{-H}^0 z \partial_z v \, dz, \end{aligned}$$

exploiting the bottom boundary condition on (u, v) . The equations for (U, V) come from vertically integrating the momentum equations (1), (2):

$$(14) \quad -fV = - \int_{-H}^0 \frac{\partial p}{\partial x} \, dz + \epsilon^2 \int_{-H}^0 \frac{\partial}{\partial z} \left(\nu \frac{\partial u}{\partial z} \right) \, dz$$

$$(15) \quad fU = - \int_{-H}^0 \frac{\partial p}{\partial y} \, dz + \epsilon^2 \int_{-H}^0 \frac{\partial}{\partial z} \left(\nu \frac{\partial v}{\partial z} \right) \, dz.$$

The depth-integrated pressure gradient in the x direction is:

$$(16) \quad \int_{-H}^0 \frac{\partial p}{\partial x} \, dz = - \frac{\partial H}{\partial x} p(x, z = -H) + \frac{\partial}{\partial x} \int_{-H}^0 p \, dz,$$

$$(17) \quad = -p_b \frac{\partial H}{\partial x} + \frac{\partial}{\partial x} \left[p_b H - \int_{-H}^0 z \frac{\partial p}{\partial z} \right],$$

$$(18) \quad = H \frac{\partial p_b}{\partial x} - \frac{\partial}{\partial x} \int_{-H}^0 z b \, dz,$$

using the Liebniz integration formula¹, where $p_b = p(x, -H(x))$ is the bottom pressure, and integration by parts². The viscous term in the x direction is:

$$(19) \quad \int_{-H}^0 \frac{\partial}{\partial z} \left(\nu \frac{\partial u}{\partial z} \right) dz = \left[\nu \frac{\partial u}{\partial z} \right]_{-H}^0 = \frac{\tau^x}{\epsilon^2} - \nu \frac{\partial u}{\partial z} \Big|_{z=-H},$$

which uses the boundary condition (34).

Therefore (14) and (15) lead to:

$$(20) \quad -fV = -H \frac{\partial p_b}{\partial x} - \frac{\partial \chi}{\partial x} - \tau_b^x + \tau^x,$$

$$(21) \quad fU = -H \frac{\partial p_b}{\partial y} - \frac{\partial \chi}{\partial y} - \tau_b^y + \tau^y,$$

where

$$(22) \quad \chi = - \int_{-H}^0 z b \, dz$$

is the baroclinic potential energy, and $\epsilon^2 \nu \partial_z(u, v)|_{z=-H} \equiv (\tau_b^x, \tau_b^y) = \tau_b$ is the bottom stress on the fluid.

Next, divide by water depth H :

$$(23) \quad -\frac{fV}{H} = -\frac{\partial p_b}{\partial x} - \frac{1}{H} \frac{\partial \chi}{\partial x} + \frac{\tau^x - \tau_b^x}{H}$$

$$(24) \quad \frac{fU}{H} = -\frac{\partial p_b}{\partial y} - \frac{1}{H} \frac{\partial \chi}{\partial y} + \frac{\tau^y - \tau_b^y}{H},$$

¹The Liebniz integration formula reads:

$$\partial_x \left(\int_{a(x)}^{b(x)} f(x, y) \, dy \right) = \partial_x b f(x, b(x)) - \partial_x a f(x, a(x)) + \int_{a(x)}^{b(x)} \partial_x f(x, y) \, dy.$$

²Integration by parts gives:

$$\begin{aligned} \int_{-H}^0 z \frac{\partial p}{\partial z} \, dz &= [zp]_{-H}^0 - \int_{-H}^0 p \, dz, \\ &= p_b H - \int_{-H}^0 p \, dz. \end{aligned}$$

and take the curl³

$$\begin{aligned}
(26) \quad & \frac{\partial}{\partial y} \left\{ -\frac{fV}{H} \right\} - \frac{\partial}{\partial x} \left\{ \frac{fU}{H} \right\} = \frac{\partial}{\partial y} \left\{ -\frac{\partial p_b}{\partial x} - \frac{1}{H} \frac{\partial \chi}{\partial x} + \frac{\tau^x - \tau_b^x}{H} \right\} \\
& - \frac{\partial}{\partial x} \left\{ -\frac{\partial p_b}{\partial y} - \frac{1}{H} \frac{\partial \chi}{\partial y} + \frac{\tau^y - \tau_b^y}{H} \right\}, \\
(27) \quad & \Rightarrow \frac{\partial}{\partial y} \left\{ -\frac{f}{H} \frac{\partial \Psi}{\partial x} \right\} + \frac{\partial}{\partial x} \left\{ \frac{f}{H} \frac{\partial \Psi}{\partial y} \right\} = -\frac{\partial}{\partial y} \left\{ \frac{1}{H} \frac{\partial \chi}{\partial x} \right\} + \frac{\partial}{\partial x} \left\{ \frac{1}{H} \frac{\partial \chi}{\partial y} \right\} - \hat{\mathbf{z}} \cdot \left(\nabla \times \frac{\tau - \tau_b}{H} \right), \\
(28) \quad & \Rightarrow -\frac{\partial}{\partial y} \left(\frac{f}{H} \right) \frac{\partial \Psi}{\partial x} + \frac{\partial}{\partial x} \left(\frac{f}{H} \right) \frac{\partial \Psi}{\partial y} = -\frac{\partial}{\partial y} \left(\frac{1}{H} \right) \frac{\partial \chi}{\partial x} + \frac{\partial}{\partial x} \left(\frac{1}{H} \right) \frac{\partial \chi}{\partial y} - \hat{\mathbf{z}} \cdot \left(\nabla \times \frac{\tau - \tau_b}{H} \right), \\
(29) \quad & \Rightarrow J \left(\frac{f}{H}, \Psi \right) = J \left(\frac{1}{H}, \chi \right) - \hat{\mathbf{z}} \cdot \left(\nabla \times \frac{\tau - \tau_b}{H} \right),
\end{aligned}$$

for streamfunction of the depth-integrated horizontal flow Ψ where

$$\begin{aligned}
(29) \quad & V = \frac{\partial \Psi}{\partial x}, \\
& U = -\frac{\partial \Psi}{\partial y},
\end{aligned}$$

and Jacobian $J(A, B) = \partial_x A \partial_y B - \partial_y A \partial_x B$. Notice how this recipe (vertically integrate, divide by H , then take the curl) eliminates the bottom pressure. The final result (28) is the same as Eq. (20) of *Peterson and Callies* (in review). It is called (a simplified form of) the **geostrophic contour vorticity balance** (GCVB) by *Waldman and Giordani* (2023), although *Peterson and Callies* (in review) call it the “barotropic vorticity equation”. It is not the same equation as used by *Nøst and Isachsen* (2003) (which *Waldman and Giordani* (2023) call the “transport divergence vorticity balance”).

Notice how the GCVB (28) is an equation for the streamfunction of the depth-integrated flow Ψ given the buoyancy field (and hence the JEBAR term) and the surface and bottom stresses. The GCVB shows how depth-integrated flow Ψ across f/H contours is driven by (i) the joint effect of baroclinicity and relief (JEBAR), (ii) curl of wind stress divided by H , (iii) curl of the bottom stress divided by H .

3. SOLUTION TO FRICTIONAL GEOSTROPHIC EQUATIONS IN STEADY STATE

Start from (1)–(2) and write the pressure as

$$(30) \quad p(x, y, z) = p_s(x, y) + \int_z^0 b(x, y, z') dz',$$

$$(31) \quad \Rightarrow \partial_x p(x, y, z) = \partial_x p_s(x, y) + \int_z^0 \partial_x b(x, y, z') dz'$$

$$(32) \quad \Rightarrow p_b(x, y) \equiv p(x, y, z = -H(x, y)) = p_s(x, y) + \int_{-H(x, y)}^0 b(x, y, z') dz',$$

³Recall that

$$(25) \quad \hat{\mathbf{z}} \cdot (\nabla \times \mathbf{U}) = \partial_x V - \partial_y U.$$

where $p_s(x, y)$ is the surface pressure (and assuming that $p_s \ll p(-H)$). Solve these frictional geostrophic equations using a Green's function, defined so that $\mathbf{u} = u + iv$, thus

$$(33) \quad \epsilon^2 \frac{\partial}{\partial z} \left(\nu \frac{\partial \mathbf{u}}{\partial z} \right) - i f \mathbf{u} = \mathbf{f},$$

$$(34) \quad \text{Surface } z = 0 : \epsilon^2 \nu \partial_z \mathbf{u} = \tau^x + i \tau^y,$$

$$(35) \quad \text{Bottom } z = -H : \mathbf{u} = 0,$$

where $\mathbf{f} = \frac{\partial p}{\partial x} + i \frac{\partial p}{\partial y}$ is the pressure gradient from (31).

The corresponding Green's function satisfies:

$$(36) \quad \epsilon^2 \frac{\partial}{\partial z} \left(\nu \frac{\partial G_u}{\partial z} \right) - i f G_u = \delta(z - \xi),$$

$$(37) \quad \text{Surface } z = 0 : \epsilon^2 \nu \partial_z G_u = 0,$$

$$(38) \quad \text{Bottom } z = -H : G_u = 0.$$

We find that the solution for constant viscosity $\nu = \nu_0$ is:

$$(39) \quad G_u(z; \xi) = \frac{i^{3/2} e^{-\sqrt{i}\varphi(z+\xi)}}{2\nu_0 \epsilon^2 \varphi \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)} \begin{cases} -e^{2\sqrt{i}\varphi \xi} + e^{2\sqrt{i}\varphi(z+H(x))} + e^{2\sqrt{i}\varphi(z+\xi+H(x))} - 1 & \text{for } z \leq \xi, \\ -e^{2\sqrt{i}\varphi z} + e^{2\sqrt{i}\varphi(\xi+H(x))} + e^{2\sqrt{i}\varphi(z+\xi+H(x))} - 1 & \text{for } z \geq \xi, \end{cases}$$

$$\Rightarrow G_u(z; 0) = \frac{i^{3/2} e^{-\sqrt{i}\varphi z}}{\nu_0 \epsilon^2 \varphi \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)} \left[e^{2\sqrt{i}\varphi(z+H(x))} - 1 \right],$$

$$(40)$$

$$\Rightarrow \partial_z G_u(z; \xi)|_{z=-H(x)} = \frac{-e^{-\sqrt{i}\varphi(\xi-H(x))}}{\nu_0 \epsilon^2 \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)} \left(1 + e^{2\sqrt{i}\varphi \xi} \right),$$

$$(41)$$

$$\Rightarrow \partial_z G_u(z; 0)|_{z=-H(x)} = \frac{-2e^{\sqrt{i}\varphi H(x)}}{\nu_0 \epsilon^2 \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)},$$

where $\varphi = \sqrt{(f/\nu_0)}/\epsilon$ (see `Example_barotropic_GeostrophicFlow_v0.2.ipynb`). And then

$$(42) \quad \mathbf{u}(z; x) = \int_{-H(x)}^0 G_u(z; \xi) [\partial_x p(x, \xi) + i \partial_y p(x, \xi)] d\xi - G_u(z; 0) (\tau^x + i \tau^y),$$

The baroclinic pressure (due to b) adds to the barotropic pressure (due to $p_s(x, y)$) in the integrand of this expression. Otherwise, everything else is known (remember that x in these expressions stands for (x, y) in general).

Also, the bottom current gradient is given by

$$(43) \quad -\partial_z \mathbf{u}(z; x)|_{z=-H(x)} = \int_{-H(x)}^0 \frac{e^{-\sqrt{i}\varphi(\xi-H(x))}}{\nu_0 \epsilon^2 \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)} \left(1 + e^{2\sqrt{i}\varphi \xi} \right) [\partial_x p(x, \xi) + i \partial_y p(x, \xi)] d\xi$$

$$- \frac{2e^{\sqrt{i}\varphi H(x)}}{\nu_0 \epsilon^2 \left(1 + e^{2\sqrt{i}\varphi H(x)} \right)} (\tau^x + i \tau^y),$$

using (40) and (41), and therefore the bottom stress τ_b on the fluid is

$$(44) \quad \begin{aligned} \tau_b = -\nu_0 \epsilon^2 \partial_z \mathbf{u}(z; x)|_{z=-H(x)} &= \int_{-H(x)}^0 \frac{e^{-\sqrt{i}\varphi(\xi-H(x))}}{1 + e^{2\sqrt{i}\varphi H(x)}} \left(1 + e^{2\sqrt{i}\varphi \xi}\right) [\partial_x p(x, \xi) + i\partial_y p(x, \xi)] d\xi \\ &\quad - \frac{2e^{\sqrt{i}\varphi H(x)}}{1 + e^{2\sqrt{i}\varphi H(x)}} (\tau^x + i\tau^y). \end{aligned}$$

Notice how the derivatives of the surface pressure part of p end up multiplying expressions that are functions of x .

4. FRICTIONAL GEOSTROPHIC EQUATIONS CLOSURE WITH THE GCVB EQUATION

The problem is closed using the GCVB equation, as follows (*Salmon* 1998, p. 170). This determines the unknown surface pressure field given the surface stress (and vice versa). In particular, work from (20)–(21),

$$(45) \quad \mathfrak{U}(z; x) \equiv U + iV = \int_{-H(x)}^0 \mathbf{u}(z; x) dz,$$

$$(46) \quad = \int_{-H(x)}^0 \left\{ \int_{-H(x)}^0 G_u(z; \xi) [\partial_x p(x, \xi) + i\partial_y p(x, \xi)] d\xi - G_u(z; 0) (\tau^x + i\tau^y) \right\} dz.$$

Computing the z integral of the Green's function $G_u(z; \xi)^4$ gives

$$(47) \quad \begin{aligned} \int_{-H(x)}^0 G_u(z; \xi) dz &= \frac{-ie^{-\sqrt{i}\varphi \xi} (e^{\sqrt{i}\varphi \xi} - e^{\sqrt{i}\varphi H(x)}) (e^{\sqrt{i}\varphi(\xi+H(x))} - 1)}{\nu_0 \epsilon^2 \varphi^2 (1 + e^{2\sqrt{i}\varphi H(x)}), \\ \Rightarrow \int_{-H(x)}^0 G_u(z; 0) dz &= \frac{-i (e^{\sqrt{i}\varphi H(x)} - 1)^2}{\nu_0 \epsilon^2 \varphi^2 (1 + e^{2\sqrt{i}\varphi H(x)})}. \end{aligned}$$

Therefore,

$$(48) \quad \begin{aligned} \mathfrak{U}(x) &= \frac{-i}{\nu_0 \epsilon^2 \varphi^2 (1 + e^{2\sqrt{i}\varphi H(x)})} \times \\ &\quad \left\{ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi \xi} (e^{\sqrt{i}\varphi \xi} - e^{\sqrt{i}\varphi H(x)}) (e^{\sqrt{i}\varphi(\xi+H(x))} - 1) [\partial_x p(x, \xi) + i\partial_y p(x, \xi)] d\xi \right. \\ &\quad \left. + (e^{\sqrt{i}\varphi H(x)} - 1)^2 (\tau^x + i\tau^y) \right\} \end{aligned}$$

(see `PlanetaryGeostrophicTheory_v0.2.ipynb`).

Eqns. (20) and (21) also imply that:

$$(49) \quad if\mathfrak{U} = -H \left(\frac{\partial p_b}{\partial x} + i \frac{\partial p_b}{\partial y} \right) - \left(\frac{\partial \chi}{\partial y} + i \frac{\partial \chi}{\partial x} \right) + \tau^x + i\tau^y - \tau_b^x - i\tau_b^y.$$

Therefore, for an unstratified fluid ($\chi = 0, \nabla p_b = \nabla p_s$), specifying the surface pressure gradient ∇p_s closes the problem. We compute the surface stress from (49) by substituting (44). Then

⁴This presumes (correctly) that the denominator in (39) does not vanish.

$u(z), v(z), U, V$ and the force balance can all be computed. See Figure 1 for an example calculation. Alternatively, the surface stress can be specified, and the pressure gradient can be computed (see `Example_barotropic_GeostrophicFlow_v0.3.ipynb`). In practice, it seems that the bottom stress always vanishes. Why?? Is this correct??

For a stratified fluid, b and hence χ (from (22)) are specified. The total pressure field $p(x, y, z)$ consists of a known buoyancy-driven piece (baroclinic) and a surface pressure driven piece from (31). Similarly, the bottom pressure field $p_b(x, y)$ consists of a known buoyancy-driven piece and a surface pressure driven piece from (32). Thus, (49) leads to:

$$(50) \quad if\mathfrak{U}(x) = - \int_{-H(x)}^0 (\partial_x + i\partial_y) p(x, z) dz + \tau_s(x) - \tau_b(x),$$

leaving the vertically-integrated pressure gradient term explicit. Substituting for $\mathfrak{U}$ from (48) and rewriting $\phi^2 = f/(\nu_0\epsilon^2)$ gives:

$$(51) \quad \frac{1}{1 + e^{2\sqrt{i}\varphi H(x)}} \left\{ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) (\partial_x + i\partial_y) p(x, \xi) d\xi \right. \\ \left. + \left(e^{\sqrt{i}\varphi H(x)} - 1 \right)^2 \tau_s(x) \right\} = \tau_s(x) - \tau_b(x) - \int_{-H(x)}^0 (\partial_x + i\partial_y) p(x, z) dz.$$

And substituting for the bottom stress τ_b from (44)

$$\frac{1}{1 + e^{2\sqrt{i}\varphi H(x)}} \left\{ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) (\partial_x + i\partial_y) p(x, \xi) d\xi \right. \\ \left. + \left(e^{\sqrt{i}\varphi H(x)} - 1 \right)^2 \tau_s(x) \right\} = \tau_s(x) - \int_{-H(x)}^0 (\partial_x + i\partial_y) p(x, z) dz \\ - \int_{-H(x)}^0 \frac{e^{-\sqrt{i}\varphi(\xi-H(x))}}{1 + e^{2\sqrt{i}\varphi H(x)}} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) (\partial_x + i\partial_y) p(x, \xi) d\xi + \frac{2e^{\sqrt{i}\varphi H(x)}}{1 + e^{2\sqrt{i}\varphi H(x)}} \tau_s(x).$$

Thus, substituting for the hydrostatic pressure from (30):

$$\frac{1}{1 + e^{2\sqrt{i}\varphi H(x)}} \times \\ \left\{ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) (\partial_x + i\partial_y) \left[p_s(x) + \int_{\xi}^0 b(x, z') dz' \right] d\xi \right. \\ \left. + \left(e^{\sqrt{i}\varphi H(x)} - 1 \right)^2 \tau_s(x) \right\} = \tau_s(x) - \int_{-H(x)}^0 (\partial_x + i\partial_y) \left[p_s(x, y) + \int_z^0 b(x, y, z') dz' \right] dz \\ - \int_{-H(x)}^0 \frac{e^{-\sqrt{i}\varphi(\xi-H(x))}}{1 + e^{2\sqrt{i}\varphi H(x)}} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) (\partial_x + i\partial_y) \left[p_s(x) + \int_{\xi}^0 b(x, z') dz' \right] d\xi \\ + \frac{2e^{\sqrt{i}\varphi H(x)}}{1 + e^{2\sqrt{i}\varphi H(x)}} \tau_s(x).$$

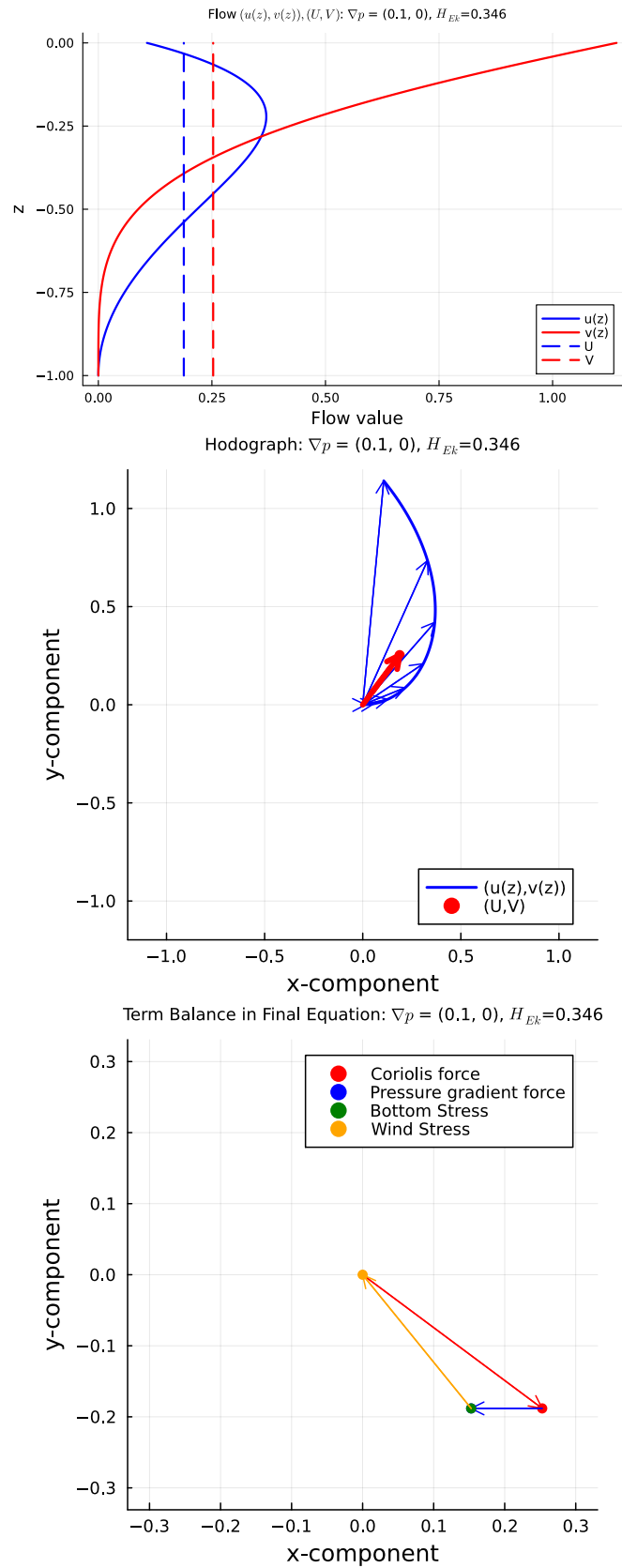


FIGURE 1. Example solution to the barotropic frictional geostrophic flow problem from `Example_barotropic_GeostrophicFlow_v0.2.ipynb`.

Rearranging to factor $\tau_s(x)$ terms on the right gives:

$$\begin{aligned}
& \frac{1}{1 + e^{2\sqrt{i}\varphi H(x)}} \times \\
& \left\{ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) (\partial_x + i\partial_y) \left[p_s(x) + \int_{\xi}^0 b(x, z') dz' \right] d\xi \right\} \\
& = \frac{4\tau_s(x)}{e^{\sqrt{i}\varphi H(x)} + e^{-\sqrt{i}\varphi H(x)}} - \int_{-H(x)}^0 (\partial_x + i\partial_y) \left[p_s(x, y) + \int_z^0 b(x, y, z') dz' \right] dz \\
& - \int_{-H(x)}^0 \frac{e^{-\sqrt{i}\varphi(\xi-H(x))}}{1 + e^{2\sqrt{i}\varphi H(x)}} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) (\partial_x + i\partial_y) \left[p_s(x) + \int_{\xi}^0 b(x, z') dz' \right] d\xi.
\end{aligned}$$

Hence, manipulating,

$$\begin{aligned}
& \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) (\partial_x + i\partial_y) \left[p_s(x) + \int_{\xi}^0 b(x, z') dz' \right] d\xi \\
& = \left(1 + e^{2\sqrt{i}\varphi H(x)} \right) \left[\frac{4\tau_s(x)}{e^{\sqrt{i}\varphi H(x)} + e^{-\sqrt{i}\varphi H(x)}} - \int_{-H(x)}^0 (\partial_x + i\partial_y) \left[p_s(x, y) + \int_z^0 b(x, y, z') dz' \right] dz \right] \\
& - \int_{-H(x)}^0 e^{-\sqrt{i}\varphi(\xi-H(x))} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) (\partial_x + i\partial_y) \left[p_s(x) + \int_{\xi}^0 b(x, z') dz' \right] d\xi.
\end{aligned}$$

Hence, breaking apart the integrals to pull the surface pressure terms out:

$$\begin{aligned}
& (\partial_x + i\partial_y) p_s(x) \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) d\xi \\
& + \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) (\partial_x + i\partial_y) \int_{\xi}^0 b(x, z') dz' d\xi \\
& = \left(1 + e^{2\sqrt{i}\varphi H(x)} \right) \left[\frac{4\tau_s(x)}{e^{\sqrt{i}\varphi H(x)} + e^{-\sqrt{i}\varphi H(x)}} - H(x) (\partial_x + i\partial_y) p_s(x, y) - \int_{-H(x)}^0 (\partial_x + i\partial_y) \int_z^0 b(x, y, z') dz' dz \right] \\
& - (\partial_x + i\partial_y) p_s(x) \int_{-H(x)}^0 e^{-\sqrt{i}\varphi(\xi-H(x))} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) d\xi \\
& - \int_{-H(x)}^0 e^{-\sqrt{i}\varphi(\xi-H(x))} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) (\partial_x + i\partial_y) \int_{\xi}^0 b(x, z') dz' d\xi.
\end{aligned}$$

Moving the surface pressure terms to the left gives:

$$\begin{aligned}
& (\partial_x + i\partial_y) p_s(x) \left\{ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) + e^{-\sqrt{i}\varphi(\xi-H(x))} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) d\xi + \right. \\
& \left. \left(1 + e^{2\sqrt{i}\varphi H(x)} \right) H(x) \right\} \\
&= \left(1 + e^{2\sqrt{i}\varphi H(x)} \right) \left[\frac{4\tau_s(x)}{e^{\sqrt{i}\varphi H(x)} + e^{-\sqrt{i}\varphi H(x)}} - \int_{-H(x)}^0 (\partial_x + i\partial_y) \int_z^0 b(x, y, z') dz' dz \right] \\
&- \int_{-H(x)}^0 e^{-\sqrt{i}\varphi(\xi-H(x))} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) (\partial_x + i\partial_y) \int_\xi^0 b(x, z') dz' d\xi \\
&- \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) (\partial_x + i\partial_y) \int_\xi^0 b(x, z') dz' d\xi
\end{aligned}$$

Manipulating the terms gives:

$$\begin{aligned}
& (\partial_x + i\partial_y) p_s(x) \left\{ \int_{-H(x)}^0 e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) + e^{-\sqrt{i}\varphi(\xi-H(x))} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) d\xi + \right. \\
& \left. \left(1 + e^{2\sqrt{i}\varphi H(x)} \right) H(x) \right\} \\
&= \frac{4 \left(1 + e^{2\sqrt{i}\varphi H(x)} \right) \tau_s(x)}{e^{\sqrt{i}\varphi H(x)} + e^{-\sqrt{i}\varphi H(x)}} - \\
&\int_{-H(x)}^0 \left[1 + e^{2\sqrt{i}\varphi H(x)} + e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H(x)} \right) \left(e^{\sqrt{i}\varphi(\xi+H(x))} - 1 \right) + e^{-\sqrt{i}\varphi(\xi-H(x))} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) \right] (\partial_x + i\partial_y) \int_\xi^0
\end{aligned}$$

Now define $A(x, \xi)$ as

$$\begin{aligned}
A(x, \xi) &\equiv e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi\xi} - e^{\sqrt{i}\varphi H} \right) \left(e^{\sqrt{i}\varphi(\xi+H)} - 1 \right) + e^{-\sqrt{i}\varphi(\xi-H)} \left(1 + e^{2\sqrt{i}\varphi\xi} \right) \\
&= e^{-\sqrt{i}\varphi\xi} \left(e^{\sqrt{i}\varphi(2\xi+H)} - e^{\sqrt{i}\varphi(\xi+2H)} - e^{\sqrt{i}\varphi\xi} + e^{\sqrt{i}\varphi H} \right) + e^{-\sqrt{i}\varphi(\xi-H)} + e^{\sqrt{i}\varphi(\xi+H)} \\
&= 2e^{\sqrt{i}\varphi(\xi+H)} - e^{2\sqrt{i}\varphi H} - 1 + 2e^{\sqrt{i}\varphi(H(x)-\xi)}, \\
(52) \quad &\implies \int_{-H}^0 A(x, \xi) d\xi = \frac{2}{\sqrt{i}\varphi} \left(e^{2\sqrt{i}\varphi H} - 1 \right) - H \left(e^{2\sqrt{i}\varphi H} + 1 \right)
\end{aligned}$$

and

$$1 + e^{2\sqrt{i}\varphi H} + A(x, \xi) = 2 \left(e^{\sqrt{i}\varphi(\xi+H)} + e^{\sqrt{i}\varphi(H-\xi)} \right).$$

Thus,

$$\begin{aligned}
& \frac{1}{\sqrt{i}\varphi} \left(e^{2\sqrt{i}\varphi H(x)} - 1 \right) (\partial_x + i\partial_y) p_s(x) \\
&= \frac{2 \left(1 + e^{2\sqrt{i}\varphi H(x)} \right) \tau_s(x)}{e^{\sqrt{i}\varphi H(x)} + e^{-\sqrt{i}\varphi H(x)}} - \int_{-H(x)}^0 \left(e^{\sqrt{i}\varphi(H(x)-z)} + e^{\sqrt{i}\varphi(z+H(x))} \right) \int_z^0 (\partial_x + i\partial_y) b(x, z') dz' dz.
\end{aligned}$$

Hence,

$$\begin{aligned} & (\partial_x + i\partial_y) p_s(x) \\ &= \frac{\sqrt{i}\varphi \tau_s(x)}{\sinh \sqrt{i}\varphi H(x)} - \frac{\sqrt{i}\varphi}{e^{2\sqrt{i}\varphi H(x)} - 1} \int_{-H(x)}^0 \left(e^{\sqrt{i}\varphi(H(x)-z)} + e^{\sqrt{i}\varphi(z+H(x))} \right) \int_z^0 (\partial_x + i\partial_y) b(x, z') dz' dz. \end{aligned}$$

To check this expression, see `PlanetaryGeostrophicTheory_v0.2.ipynb`.

This final expression is of the form (now with both (x, y) spatial arguments)

$$(53) \quad (\partial_x + i\partial_y) p_s(x, y) = F(x, y),$$

where

$$(54) \quad F(x, y) = \frac{\sqrt{i}\varphi \tau_s(x, y)}{\sinh \sqrt{i}\varphi H(x, y)} - \frac{\sqrt{i}\varphi}{e^{2\sqrt{i}\varphi H(x, y)} - 1} \int_{-H(x, y)}^0 \left(e^{\sqrt{i}\varphi(H(x, y)-z)} + e^{\sqrt{i}\varphi(z+H(x, y))} \right) \int_z^0 (\partial_x + i\partial_y) b(x, y, z') dz' dz.$$

The formal solution of (53) is

$$(55) \quad p_s(x, y) = \int_0^{T(x, y)} F(x - t, y - it) dt,$$

or, equivalently (because p_s is real)

$$(56) \quad \nabla^2 p_s(x, y) = \Re [(\partial_x - i\partial_y) F(x, y)].$$

The boundary condition is that p_s is a constant on the boundary and that the global integral of p_s vanishes.

Notice the solvability condition derived by integrating along a closed path, i.e.,

$$(57) \quad \oint_{\Gamma(x(s), y(s))} (\partial_x + i\partial_y) p_s(x, y) ds = \oint_{\Gamma(x(s), y(s))} F(x, y) ds$$

for circuit $\Gamma(x(s), y(s))$, where s measures distance along the circuit. The left hand side of this expression vanishes by the Cauchy integral theorem for holomorphic (differentiable) function $(\partial_x + i\partial_y) p_s(x, y)$. Moreover, the circuit Γ is arbitrary, which implies that

$$(58) \quad \oint_{\Gamma(x(s), y(s))} F(x, y) ds = 0 \quad \forall \Gamma,$$

which means that the circulation of F along Γ vanishes. Therefore, the curl of F must vanish everywhere:⁵

$$(60) \quad (\partial_y - i\partial_x) F(x, y) = 0 \quad \forall (x, y).$$

In words, the curl of the surface stress term in (54) must equal the curl of the buoyancy gradient term in (54) for the solution to satisfy the original (steady, linear) momentum equations (1)–(2).

5. EXAMPLE SOLUTIONS: WORK IN PROGRESS

Next steps are to solve for:

- (1) $b = 0$, $H(x, y) = 1 - x^2 - y^2$ in a circular parabolic basin. See `ExampleSolution_v0.1.ipynb`.
- (2) $b(x, y)$ given by solving the vertical diffusive buoyancy equation (see section 6) and $H(x, y) = 1 - x^2 - y^2$ in a circular parabolic basin.

6. SOLUTION TO LINEAR BUOYANCY EQUATION IN STEADY STATE

At least for steady state and negligible buoyancy advection, the buoyancy equation (8) reads:

$$(61) \quad \epsilon^2 \frac{\partial}{\partial z} \left(\kappa \frac{\partial b}{\partial z} \right) + \gamma (B(z) - b) = 0,$$

$$(62) \quad \text{Surface } z = 0 : b = 0$$

$$(63) \quad \text{Bottom } z = -H : \frac{\partial b}{\partial z} = \beta$$

for buoyancy profile $B(z)$ (in principle, B could depend on (x, y) too) and relaxation coefficient γ , and vertical buoyancy gradient at the sea bed β (which may be zero). The internal buoyancy source allows for steady-state solutions that have non-trivial solutions (the $t \rightarrow \infty$ solution to (8)–(10) is vanishing b everywhere; the *Peterson and Callies* in review solutions are transient, relaxing to zero). This problem (61) can be solved exactly using a Green's function, at least for constant diffusivity, and I suspect also for diffusivity given by (11).

In detail: Solve the Green's function problem first:

$$(64) \quad \frac{\epsilon^2}{\gamma} \frac{\partial}{\partial z} \left(\kappa \frac{\partial G_b}{\partial z} \right) - G_b = -\delta(z - \xi),$$

$$(65) \quad \text{Surface } z = 0 : G_b = 0,$$

$$(66) \quad \text{Bottom } z = -H : \frac{\partial G_b}{\partial z} = 0.$$

⁵Note Morera's theorem says that $F(x, y)$ must be holomorphic because of (58). Therefore, $F(x, y)$ must satisfy the Cauchy-Riemann equations:

$$(59) \quad \begin{aligned} \partial_x \Re\{F\} - \partial_y \Im\{F\} &= 0, \\ \partial_y \Re\{F\} + \partial_x \Im\{F\} &= 0. \end{aligned}$$

Then, writing F in terms of real and imaginary parts $F = \Re\{F\} + i\Im\{F\} \equiv F_1 + iF_2$, gives:

$$(\partial_y - i\partial_x) F = (\partial_y F_1 + \partial_x F_2) + i(\partial_y F_2 - \partial_x F_1) = 0$$

from (59). Notice how

$$\begin{aligned} \partial_y F_1 + \partial_x F_2 &\Leftrightarrow \nabla \cdot \mathbf{F}, \\ \partial_y F_2 - \partial_x F_1 &\Leftrightarrow \nabla \times \mathbf{F}, \end{aligned}$$

for vector $\mathbf{F} = (F_2; F_1)$.

This is a second order linear ordinary differential equation with two boundary conditions, which can be solved exactly. The solution for constant diffusivity $\kappa = \kappa_0$ is:

$$(67) \quad G_b(z; \xi) = \frac{e^{\phi(z+\xi)}}{2\kappa_0\phi(e^{2\phi H(x)} + 1)} \begin{cases} (1 - e^{-2\phi\xi}) \left(e^{2\phi H(x)} + e^{-2\phi z} \right) & \text{for } z \leq \xi, \\ (1 - e^{-2\phi z}) \left(e^{2\phi H(x)} + e^{-2\phi\xi} \right) & \text{for } z \geq \xi, \end{cases}$$

with $\phi = \sqrt{\gamma/\kappa_0} \epsilon$. Therefore, given $G_b(z; \xi)$,

$$(68) \quad b(z; x) = - \int_{-H(x)}^0 G_b(z; \xi) B(\xi; x) d\xi.$$

See `DiffusionEquationSolution.ipynb` and `FindBuoyancyField_and_GeostrophicFlow_v0.?.ipynb`.

In solving the momentum equations, one needs horizontal derivatives of the buoyancy field to compute the pressure gradient (31). They are:

$$(69) \quad \begin{aligned} -\partial_x b(z; x) &= \partial_x H(x) G_b(z; -H(x)) B(-H(x); x) \\ &\quad + \int_{-H(x)}^0 \partial_x G_b(z; \xi) B(\xi; x) d\xi \\ &\quad + \int_{-H(x)}^0 G_b(z; \xi) \partial_x B(\xi; x) d\xi, \end{aligned}$$

(and similarly for the y derivative—the y dependence in b , H , and B is suppressed here for simplicity). For constant κ , $G_b(z; -H(x))$ is a relatively straightforward expression:

$$(70) \quad G_b(z; -H(x)) = -(\kappa_0\phi)^{-1} \cosh(\phi(z + H(x))) \tanh(\phi H(x))$$

and $\partial_x G_b(z; \xi)$ is:

$$(71) \quad \partial_x G_b(z; \xi) = \frac{(-e^{2z\phi} - e^{2\xi\phi} + e^{2\phi(z+\xi)} + 1) e^{\phi(-z-\xi+2H(x))}}{\kappa_0 (e^{4\phi H(x)} + 2e^{2\phi H(x)} + 1)} \partial_x H(x)$$

(see `FindBuoyancyField_and_GeostrophicFlow_v0.4.ipynb`).

REFERENCES

- Klinger, B. A., and T. W. N. Haine (2019), *Ocean Circulation in Three Dimensions*, 1st ed., Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.
- Nøst, O. A., and P. E. Isachsen (2003), The large-scale time-mean ocean circulation in the Nordic Seas and Arctic Ocean estimated from simplified dynamics, *J. Mar. Res.*, *61*(2), 175–210, doi: 10.1357/002224003322005069.
- Peterson, H. G., and J. Callies (2023), Coupling between abyssal boundary layers and the interior ocean in the absence of along-slope variations, *J. Phys. Oceanogr.*, *53*(1), 307–322, doi: 10.1175/jpo-d-22-0082.1.
- Peterson, H. G., and J. Callies (in review), Connecting the local response to abyssal mixing to the basin-scale circulation, *J. Phys. Oceanogr.*
- Salmon, R. (1998), *Lectures on Geophysical Fluid Dynamics*, Oxford University Press, doi: 10.1093/oso/9780195108088.001.0001.
- Vallis, G. K. (2006), *Atmospheric and Oceanic Fluid Dynamics: Fundamentals and Large-scale Circulation*, 1st ed., 745 pp., Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, doi:10.1017/cbo9780511790447.

Waldman, R., and H. Giordani (2023), Ocean barotropic vorticity balances: Theory and application to numerical models, *J. Advances Model. Earth Sys.*, *15*(4), doi:10.1029/2022ms003276.